

AI in Geological Modeling: Research Repository Expansion Guide

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This repository documents the most valuable additions to advance AI-driven geological modeling research, organized by category. Each section contains actionable research directions, known data points, and open questions.

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🧪 Technical Deep-Dives

1.1 Computational Benchmarks

Current Gap: Most performance claims in the literature are theoretical or derived from small-scale experiments. Production-scale GPU memory footprints, training times, and inference latencies are rarely reported.

Required Benchmarks:

Technology	Metric	Current State	Target
PINNs	GPU Memory (24GB vs 48GB vs 80GB)	Mostly 2D/3D toy problems	500×500×500 voxel reservoir
Diffusion Models	Inference time per 256 ³ volume	Not standardized	<30s on A100
GNNs	Training time per 10k nodes	Varies wildly	Standardized graph sizes
Transformers	Attention memory scaling	O(n ²) theoretical	Measured on 1M tokens
NeRF/Plenoxels	Render time per view	Academic benchmarks	Production drillhole density

Standardized Test Conditions:

- **Hardware baselines:** NVIDIA A100 80GB, H100 80GB, RTX 4090 (consumer reference)
- **Dataset sizes:** 1k, 10k, 100k drillholes; 1GB, 10GB, 100GB seismic volumes

- **Voxel resolutions:** 128³, 256³, 512³, 1024³
- **Batch configurations:** Single GPU, multi-GPU (2/4/8), distributed

Deliverable: A reproducible benchmark suite with `benchmark.yml` configs and automated reporting.

1.2 Failure Modes and Edge Cases

Critical Research Gap: Literature overwhelmingly reports success cases. Understanding when AI *violates* geological principles is essential for trust and regulatory acceptance.

Technology	Known Failure Mode	Severity	Detection Method
PINNs	Violation of mass conservation when λ_{physics} is too low	High	Residual monitoring
PINNs	Spectral bias—fails on high-frequency geological boundaries	High	Fourier analysis of residuals
Diffusion Models	Generates geologically impossible cross-cutting relationships	High	Topological validation
Diffusion Models	Stratigraphic inversions (younger layers below older)	High	Age constraint checking
GNNs	Misses deposits when graph connectivity is sparse	Medium	Sensitivity analysis on edge threshold
GNNs	Over-smoothing in deep layers loses local structure	Medium	Layer-wise feature variance
CNNs/ Segmentation	Hallucinates ore shoots in homogeneous host rock	Medium	Uncertainty quantification
Transformers	Position bias—attends to drillhole spacing not geology	Medium	Attention visualization

Recommended Documentation Format:

```
Failure Case ID: PINN-001
Technology: Physics-Informed Neural Networks
Trigger:  $\lambda_{\text{physics}} < 0.01$  in 3D Darcy flow
Symptom: Negative permeability values in low-gradient regions
Root Cause: PDE residual dominates over data fidelity term
Mitigation: Adaptive loss weighting (NTK-based or gradient statistics)
Reference: [Paper/Internal Test]
```

1.3 Data Requirements Specification

Current Problem: “Pre-train on 10,000 synthetic models” is insufficiently specified for procurement or experimental design.

Required Specifications by Technology:

PINNs

Parameter	Minimum	Recommended	Production
Voxel dimensions	64 ³	256 ³	512 ³
Boundary condition points	1,000	10,000	100,000
Collocation points	100,000	1,000,000	10,000,000
PDE formulation complexity	Single physics	Coupled (flow+transport)	Multi-phase + geomechanics

Diffusion Models

Parameter	Minimum	Recommended	Production
Training samples	1,000	10,000	100,000
Sample resolution	128 ³	256 ³	512 ³
Conditioning channels	1 (elevation)	3 (drillholes, faults, topography)	10+ (multi-physics)
Seismic trace density	1 trace / 100m	1 trace / 25m	1 trace / 10m

GNNs

Parameter	Minimum	Recommended	Production
Number of drill holes	50	500	5,000
Nodes per graph	1,000	10,000	100,000
Edge connectivity	k=4 nearest neighbors	k=8 + geological distance	Adaptive threshold
Feature dimensions per node	3 (x,y,z,grade)	10 (geochemistry + geophysics)	50+ (multi-sensor)

Satellite/Remote Sensing

Parameter	Minimum	Recommended	Production
Scene count	100	1,000	10,000
Resolution (Sentinel-2)	10m	10m	10m
Spectral bands	RGB	10 bands	13 bands + SAR
Temporal coverage	Single date	1 year	5+ years

1.4 Hyperparameter Sensitivity

High-Impact Parameters (“Knobs That Matter”):

Technology	Parameter	Working Range	Breaking Range	Notes
PINNs	__physics (PDE weight)	0.1 -10.0	<0.01 or >100	Adaptive strategies preferred
PINNs	Network depth	4 -8 layers	<3 or >12	Deeper networks show spectral bias
Diffusion	Guidance scale (w)	1.0 -4.0	>10.0	Higher = more realistic but less diverse
Diffusion	Number of denoising steps	50 -250	<20	Quality vs. speed tradeoff
Diffusion	Conditioning dropout	0.1 -0.3	>0.5	Prevents over-reliance on sparse data
GNN	Attention heads	4 -8	1 or >16	Diminishing returns after 8
GNN	Message passing steps	3 -6	>10	Over-smoothing risk
GNN	Learning rate	1e-3 -1e-4	>1e-2	Very sensitive for sparse geological graphs
Transformer	Context window	2,048 -8,192	>32k (memory)	Geological sequences are long-range
Transformer	Positional encoding	Sinusoidal	Learned	Geospatial coordinates need special handling

Recommended: Publish sensitivity analysis as parallel coordinate plots and ablation studies.

Industry & Market Intelligence

2.1 Pricing Intelligence

Current Status: Limited public pricing data. Most vendors operate on quote-based enterprise sales.

Seequent (Bentley Subsurface)

Product	Pricing Model	Rate (USD)	Monthly Cap
Leapfrog Geo	Consultants Daily	\$156/day	\$2,340 (15 days)
Leapfrog Energy	Consultants Daily	\$267/day	\$4,005 (15 days)
Leapfrog Edge	Extension Daily	\$66.50/day	\$997.50 (15 days)
Geophysics	Extension Daily	\$41.25/day	\$618.75 (15 days)
Extension			
Hydrogeology	Extension Daily	\$41.25/day	\$618.75 (15 days)
Extension			
ioGAS Link	Extension Daily	\$15.00/day	\$225.00 (15 days)
Maptek Link	Extension Daily	\$6.50/day	\$97.50 (15 days)
Activation Fee	One-time	\$400	N/A

Source: Seequent official pricing page, 2026 [cite: web_search:1#0]

Subscription Types Available:

- **Shared:** Floating licenses for teams
- **Named:** Dedicated user licenses
- **Consultants Daily:** Pay-per-use for variable workloads
- **Daily Seats:** 24-hour access periods, unlimited concurrent users

Maptek Vulcan

- **Pricing Model:** Quote-based; not publicly disclosed
- **Options:** One-time purchase or subscription-based
- **Contact:** Sales team required for all pricing [cite: web_search:1#5]

VRIFY

- **Pricing Model:** Not publicly disclosed
- **Funding:** \$12.5M Series B (February 2025) for DORA AI-assisted mineral discovery platform [cite: web_search:1#3]
- **Status:** Enterprise SaaS; likely quote-based

GeologicAI

- **Pricing Model:** Not publicly available
- **Service:** Core scanning and AI analysis
- **Status:** Private company; pricing by negotiation

Research Action Items:

1. Submit RFQ to Maptek for Vulcan subscription pricing (1-year, 5-seat)
2. Request VRIFY pricing deck via sales contact
3. Contact GeologicAI for scanning rates per meter and AI analysis fees
4. Survey 10 mid-tier mining companies on actual annual software spend

2.2 Customer Interview Synthesis

Target Personas:

Role	Organization Type	Key Pain Points	AI Adoption Barriers
Exploration Geologist Resource Geologist Mine Geologist VP Exploration Data Scientist	Junior mining company	Manual wireframing is slow; data integration painful	Trust in AI-generated structures; liability concerns
	Major (BHP, Rio Tinto)	Model updates take weeks; version control chaos	Regulatory acceptance; QP sign-off requirements
	Underground operation	Daily model updates needed; limited compute	Real-time inference requirements; data connectivity
	Mid-tier producer	Budget pressure; need faster target generation	ROI uncertainty; integration with legacy workflows
	In-house AI team	Geological domain knowledge gap; messy data	Communication with geologists; data quality issues

Recommended Interview Protocol:

1. **Screening:** 5-10 working geologists, mix of majors and juniors
2. **Format:** 45-minute structured interviews + follow-up
3. **Questions:**
 - Current tool stack and annual spend
 - Time spent on manual tasks vs. interpretation
 - Biggest frustration with current 3D modeling workflow
 - What would make you trust an AI-generated geological model?
 - What would make you refuse to use AI entirely?
 - Priority stack: speed, accuracy, compliance, cost, integration

Deliverable: Synthesis report with persona profiles, pain point prioritization matrix, and feature request frequency analysis.

2.3 Regulatory Precedent

Critical Gap: No confirmed public cases of AI-assisted JORC or NI 43-101 reports accepted by regulators.

Regulatory Framework Overview

Standard	Jurisdiction	Governing Body	AI Relevance
JORC Code	Australia, New Zealand, ASX	Australasian Joint Ore Reserves Committee	Mineral Resource reporting
NI 43-101	Canada, TSX	Canadian Securities Administrators (CSA)	Technical report disclosure
SAMREC	South Africa, JSE	South African Mineral Resource Committee	Resource/reserve reporting
PERC	Europe	Pan-European Reserves & Resources Committee	Harmonized reporting
CBRR	China	Ministry of Natural Resources	Chinese reporting standard

Key Differences:

- **NI 43-101** requires a **Qualified Person (QP)** with specific education/experience credentials [cite: web_search:1#12]
- **JORC Code** requires a **Competent Person (CP)** recognized by a professional organization
- Both are often considered interchangeable for dual-listed entities [cite: web_search:1#7]
- NI 43-101 technical reports may be subject to review by securities regulatory authorities [cite: web_search:1#9]

Open Questions for Regulatory Research:

1. Has any TSX-listed company filed an NI 43-101 where AI assisted in variogram modeling or domain boundary definition?
2. What was the QP's statement regarding AI involvement?
3. Did the regulator (CSA/OSC) raise questions about AI methodology?
4. Are there any written guidance documents on AI/ML in mineral resource estimation?
5. What is the liability framework if an AI-assisted model is later found materially incorrect?

Research Strategy:

- Search SEDAR/ASX filings for keywords: “machine learning,” “artificial intelligence,” “algorithmic,” “automated interpretation”
- Contact CSA directly for guidance requests
- Interview 3-5 QPs/CPs on their current stance toward AI tools

2.4 Competitive Response Scenarios

Seequent (Bentley Systems) Current AI Position:

- **Seequent Evo** launched March 2025: Cloud-based platform with open APIs, opt-in AI/ML functionality [cite: web_search:1#17]
- **PDAC 2026 Announcements:**
- AI-driven structural insights via machine learning and computer vision [cite: web_search:1#1]
- Partnership with Orica Digital Solutions for drill-to-model workflows [cite: web_search:1#1]
- Leapfrog Geo desktop and cloud updates [cite: web_search:1#2]
- **Architecture:** Evo provides centralized data + AI-ready foundation; desktop tools (Leapfrog) enhanced with cloud compute

Likely Response to Full AI Integration:

- **Strengths:** Massive installed base, regulatory trust, deep domain integration
- **Likely Architecture:** AI as augmentation layer within Evo; not replacement of QP judgment
- **Patent Portfolio:** Unknown; requires IP search on “geological modeling AI” patents assigned to Bentley/Seequent
- **Risk:** Slow to disrupt own cash-cow licensing model

Maptek

- **Position:** Strong in mine design and production; less in exploration AI
- **Likely Response:** Acquisition or partnership with AI startup; API integration with Vulcan
- **Patent Portfolio:** Requires investigation

VRIFY

- **Position:** Pure-play AI mineral discovery; \$12.5M Series B [cite: web_search:1#8]
- **Product:** DORA (AI-assisted mineral discovery)
- **Threat:** Most direct competitor if they expand from target generation to full 3D modeling

Scenario Planning Matrix:

Scenario	Probability	Impact	Recommended Response
Seequent announces AI auto-modeling in Leapfrog	Medium	High	Differentiate on commodity-specific models
Maptek acquires AI startup	Medium	Medium	Accelerate partnership discussions
VRIFY raises Series C (\$50M+)	Medium	Medium	Lock in early customers before scaling
Regulator bans AI in resource reports	Low	High	Build regulatory compliance as core competency

Table 11 – continued

Scenario	Probability	Impact	Recommended Response
Open-source geological AI toolkit emerges	High	🟡 Medium	Contribute to community; monetize enterprise features

💰 Business & Strategy

3.1 Detailed Financial Model

Beyond TAM/SAM: Required Unit Economics

Revenue Model Options

Model	Description	Pros	Cons
SaaS Subscription	Monthly/annual per-seat or per-project	Predictable revenue; scalable	High churn risk; long sales cycles
Usage-Based	Per model, per voxel, per drillhole processed	Aligns with value; low barrier	Revenue lumpy; hard to predict
Professional Services	Custom model building + training	High margin; deep customer relationships	Not scalable; talent bottleneck
Hybrid	Base SaaS + usage overages + services	Balanced predictability and growth	Complex pricing; customer confusion

Required Metrics

Metric	Definition	Target Benchmark
Customer Acquisition Cost (CAC)	Sales + marketing spend / new customers	<\$50K for enterprise mining
Lifetime Value (LTV)	ARPU × gross margin × customer lifetime	>3× CAC
LTV:CAC Ratio	Standard SaaS health metric	>3:1
Churn by Segment	Monthly logo churn %	<5% annual for enterprise
Sales Cycle Length	First contact to signed contract	6-12 months for major mining
Professional Services Margin	(Revenue - COGS) / Revenue	>40%
Net Revenue Retention	(Starting ARR + expansion - churn) / Starting ARR	>120%
Payback Period	CAC / monthly gross margin	<18 months

Model Structure:

Year 1-3 Assumptions:

- Initial customers: 5 pilot programs
- Average contract value (ACV): \$150K/year
- Expansion revenue: 30% YoY from existing customers
- Services attach rate: 60% of new deals
- Headcount: 15 → 40 → 80
- Burn rate: \$2M → \$5M → \$8M/year

3.2 Partnership Landscape

Build vs. Buy vs. Partner Matrix

Capability	Build	Buy	Partner	Partner Candidates
Cloud Infrastructure	✗	✗	✓	AWS (SageMaker), Azure (ML Studio), GCP (Vertex AI)
GPU Compute	✗	✗	✓	NVIDIA (Omniverse, CUDA), CoreWeave, Lambda Labs
Satellite Data	✗	✗	✓	Sentinel Hub (Copernicus), Planet Labs, Maxar
Legacy Software APIs	✗	✗	✓	Seequent Evo API, Maptek Vulcan SDK, Micromine API
Drillhole Data	✗	✗	✓	WAMEX, NSGD, government geological surveys
Domain Expertise	✗	✓	✓	Hire ex-BHP/Rio geologists; partner with universities
AI Core (Diffusion/GNN)	✓	✗	✗	Internal R&D team
Regulatory Compliance	✓	✗	✓	Consulting firms (SRK, AMC, CSA Global)

Partnership Priorities:

1. **Immediate:** Cloud provider credits (startup programs: AWS Activate, Azure for Startups)
2. **Short-term:** Satellite data API integration (Sentinel Hub has free tier)
3. **Medium-term:** Legacy software interoperability (Seequent Evo API is new and open)
4. **Long-term:** Co-development with major mining company (joint venture for data access)

3.3 Talent Map

Target Profiles: The 20 People Who Could Build This

Category	Known Individuals/Teams	Location	Expertise
Geological AI Research	Zhejiang Lab GeoGPT team	Hangzhou, China	LLMs for geology
GNN for Mineral Exploration	KADM authors (Korea)	South Korea	Knowledge-aware deep mining
Attention Mechanisms	KDCGAT researchers	Various	Knowledge-driven contextual GAT
Industry Veterans	ex-BHP AI team members	Melbourne/Perth	Production ML in mining
Visualization	ex-Rio Tinto RTVis engineers	Various	Real-time 3D geological visualization
PINNs for Geoscience	Prof. Kamyar Azizzadenesheli (Purdue)	USA	Physics-informed ML
Diffusion Models	Stability AI / OpenAI alumni	Various	Generative modeling
Geostatistics + ML	Prof. Jef Caers (Stanford)	USA	Geostatistical learning
Structural Geology AI	Seequent AI team (Alexander Wilson)	Toronto/Christchurch	ML for structural interpretation
Remote Sensing	Prof. Ribana Roscher (Jülich)	Germany	Deep learning for Earth observation

Recruitment Strategy:

- **Tier 1:** Full-time hires from major mining company AI teams (high comp, but domain expertise is irreplaceable)
- **Tier 2:** Academic partnerships (PhD co-supervision, research grants)
- **Tier 3:** Advisory board (2-3 renowned professors for credibility and network)

3.4 Technical Debt Assessment

Lock-in Risks and Migration Paths

Current Choice	Lock-in Risk	Migration Path	Trigger
PyTorch	Medium	JAX, TensorFlow	If PyTorch ecosystem declines
Diffusion Models	High	Flow Matching (Rectified Flow), Consistency Models	If sampling speed becomes critical
Transformers	Medium	State Space Models (Mamba, RWKV), Linear Attention	If $O(n^2)$ scaling blocks context windows

Table 16 – continued

Current Choice	Lock-in Risk	Migration Path	Trigger
CUDA	High	ROCm, oneAPI, Triton	If NVIDIA pricing/availability changes
Specific GNN Library	Medium	PyG → DGL → Custom	If library maintenance stops
Cloud Provider	Medium	Multi-cloud (Kubernetes)	If single provider pricing increases

Mitigation Strategies:

- 1. **Abstraction layers:** Wrap PyTorch models in framework-agnostic interfaces
- 2. **Model export:** Standardize on ONNX for deployment; avoid framework-specific ops
- 3. **Modular architecture:** Swappable components (diffusion flow matching; transformer Mamba)
- 4. **Cloud-agnostic:** Kubernetes + Terraform for multi-cloud portability

 Validation & Evidence

4.1 Reproducible Benchmarks

Dataset 1: WAMEX (Western Australia) Overview:

- **Repository:** Western Australian Mineral Exploration reports (WAMEX) [cite: web_search:1#11]
- **Size:** 120,000+ reports; 105,000+ publicly available with raw data
- **Data Types:** Drilling data, geochemistry results, 3D models, gravity surveys, structural maps
- **Access:** Free via WAMEX search portal; bulk download available on hard drive from Perth/Kalgoorlie [cite: web_search:1#14]
- **Confidentiality:** 5-year embargo under Mining Act 1978 [cite: web_search:1#10]
- **Update Cycle:** May and November each year [cite: web_search:1#14]

Known Data Quality Issues:

- WAMEX geochemical data may contain spurious results (unit reporting errors, incorrect analyte assignment) [cite: web_search:1#6]
- Automated compilation loses ~15% of data; low/high cuts remove another 7-20% (including valid data) [cite: web_search:1#6]
- Multiple analytical techniques combined in same table—not all directly comparable [cite: web_search:1#6]
- Historical gravity data requires rigorous harmonization due to varying contractors, methods, and datums [cite: web_search:1#15]

Benchmark Protocol:

1. Select 5 open-file reports with complete drillhole data + assay results
2. Define task: 3D grade interpolation from drillholes only
3. Run baseline: Ordinary Kriging (OK) in commercial software (Leapfrog, Micromine)
4. Run AI methods: PINN, GNN, Diffusion-based infill
5. Metrics: RMSE, MAE, correlation coefficient at cross-validation locations
6. Compute cost: GPU hours, training time, inference time per model
7. Publish: Code, data, results on GitHub with DOI

Dataset 2: NSGD (Nevada Geothermal) **Status:** Search did not return specific open dataset information. Requires further investigation.

Alternative Open Datasets:

- USGS National Geologic Map Database
- Geoscience Australia (GA) datasets
- British Geological Survey Open Data
- Natural Resources Canada Geospatial Data

Deliverable: Public benchmark leaderboard with standardized train/test splits and evaluation scripts.

4.2 Blind Test Protocol

Objective: Measure geologist trust, actionability, and perceived quality of AI-generated models without disclosure of AI origin.

Protocol Design:

Phase 1: Model Generation

- Select 5 real deposits with known geology (but not widely published)
- Generate 3D models using: (a) Human expert, (b) AI method A, (c) AI method B
- Render consistent visualizations (same color scheme, orientation, scale)

Phase 2: Blind Evaluation

- Recruit 20 geologists (mix of experience levels)
- Randomize model presentation order
- Ask evaluators to score each model on:
 1. Geological plausibility (1-10)
 2. Structural consistency (1-10)
 3. Usefulness for drill planning (1-10)
 4. Trustworthiness for resource estimation (1-10)
 5. Overall quality vs. their own work (1-10)
- Open comment: "What concerns you about this model?"

Phase 3: Disclosure & Follow-up

- Reveal which models were AI-generated
- Measure change in trust scores (before/after disclosure)

- Identify specific features that triggered suspicion or confidence

Ethical Considerations:

- IRB approval required for human subjects research
- Debriefing must include full disclosure of AI involvement
- No evaluative consequences for participants (not a hiring test)

4.3 Economic Validation

Case Study Template:

```
## Case Study: [Deposit Name] — [Commodity]

### Project Context
- Operator: [Company]
- Location: [Jurisdiction]
- Deposit Type: [Porphyry/IOCG/etc.]
- Exploration Budget: $[X]M
- Timeline: [Years]

### Counterfactual Analysis
"If we had [AI Platform] for the [Project Name]:"

| Metric | Without AI (Actual) | With AI (Simulated) | Delta |
|-----|-----|-----|-----|
| Drill meters required | 50,000m | 35,000m | -30% |
| Drill cost (@$150/m) | $7.5M | $5.25M | **-$2.25M** |
| Modeling person-weeks | 40 weeks | 10 weeks | -75% |
| Modeling cost (@$5K/week) | $200K | $50K | **-$150K** |
| Time to first resource | 24 months | 18 months | -6 months |
| Opportunity cost (discount) | $[X]M | $[Y]M | **-$[Z]M** |

### Key Assumptions
- AI model accuracy within 15% of final resource
- Drilling reduction achieved through better targeting
- No additional data collection costs

### Validation Required
- [ ] Operator interview to confirm actual costs
- [ ] Drill plan comparison (actual vs. AI-recommended)
- [ ] Model accuracy assessment vs. subsequent drilling
```

Domain-Specific Extensions

5.1 Commodity-Specific Models

Rationale: A generic platform is less valuable than commodity-specialized models. Each commodity has distinct geological controls, data types, and modeling priorities.

Commodity	Deposit Type	Key Geological Controls	Priority Data	AI Opportunities
Copper	Porphyry	Alteration zonation, stockwork density, sulfide assemblage	Hyperspectral, geochemistry, magnetics	Alteration classification from drill core imagery
Copper	IOCG	Structural control, breccia zones, iron oxide alteration	Gravity, magnetics, EM	Structural control prediction from geophysics
Gold	Orogenic	Shear zones, alteration halos, sulfide association	Structural maps, geochemistry, gravity	Shear zone prediction from sparse drill data
Gold	Carlin	Host rock chemistry, realgar/stibnite, decarbonatization	Geochemistry, hyperspectral, seismic	Host rock prediction from regional geochemistry
Lithium	Pegmatite	Spodumene abundance, tantalite association, fractionation	Handheld XRF, hyperspectral, magnetics	Pegmatite texture classification from core photos
Lithium	Brine	Aquifer geometry, evaporation rate, source rock leaching	Geophysics (TEM/MT), isotopes, well logs	Brine aquifer prediction from EM data
REE	Carbonatite	REE mineralogy, weathering profile, gangue composition	Geochemistry, mineralogy, radiometrics	REE grade prediction from portable XRF
Nickel	Magmatic Sulfide	Sulfide fraction, chalcopyrite/pentlandite ratio, PGE	Geochemistry, petrology, EM	Sulfide texture classification
Uranium	Unconformity	Graphite basement, alteration clay, redox boundary	EM, gravity, radiometrics, geochemistry	Basement graphitic unit prediction
Zinc-Lead	VMS	Volcanic stratigraphy, hydrothermal alteration, stockwork	Geochemistry, magnetics, EM	Volcanic facies prediction from geophysics

Recommended Approach:

- Start with **2-3 commodities** (e.g., Porphyry Cu, Orogenic Au, Lithium Pegmatite)
- Develop commodity-specific loss functions and geological constraints
- Build training datasets from WAMEX/NSGD filtered by deposit type
- Partner with commodity specialists for validation

5.2 Jurisdiction-Specific Regulations

Compliance Requirements by Jurisdiction

Standard	Jurisdiction	Key Requirements	AI Implications
JORC Code 2012	Australia, New Zealand, ASX	Competent Person (CP) sign-off; transparent assumptions; materiality	AI can assist but CP must take responsibility; need explainability
NI 43-101	Canada, TSX	Qualified Person (QP); specific education + experience; filing on SEDAR	QP must supervise AI; methodology disclosure required
SAMREC 2016	South Africa, JSE	Competent Person; reasonable prospects for eventual economic extraction	Similar to JORC; AI as tool, not substitute for judgment
PERC 2021	Europe	Harmonized with CRIRSCO; public reporting of exploration results	Emerging standard; less precedent on AI
CBRR	China	Ministry of Natural Resources oversight; resource reserve classification	Rapidly evolving; potential for AI-first regulation

Research Deliverables:

1. Side-by-side comparison matrix of AI disclosure requirements
2. Template “AI Methods Statement” for inclusion in technical reports
3. Precedent search: Has any AI/ML language appeared in filed NI 43-101 or JORC reports?
4. Interview 3 QPs/CPs on their willingness to sign off AI-assisted work

5.3 Climate and ESG Integration

Growth Markets with Regulatory Tailwinds

Application	Market Driver	Data Types	AI Opportunity
Carbon Sequestration Site Characterization	CCS mandates; 45Q tax credits	Seismic, well logs, caprock integrity	Reservoir seal prediction; injection plume monitoring
Critical Mineral Supply Chain Mapping	IRA, EU Critical Raw Materials Act	Satellite, trade data, geological surveys	Prospectivity mapping + supply chain risk
Tailings Dam Monitoring	Global Tailings Standard; investor pressure	InSAR, geotechnical sensors, drone imagery	Failure prediction; deformation anomaly detection

Table 19 – continued

Application	Market Driver	Data Types	AI Opportunity
Mine Closure Planning	Progressive closure bonds; regulatory requirements	Topography, hydrology, geochemistry, vegetation	Closure scenario optimization; long-term monitoring
Geothermal Resource Assessment	Renewable energy targets	Temperature gradients, geophysics, well tests	Resource temperature prediction; drilling target optimization

Strategic Value: These applications have shorter regulatory paths and less liability risk than Mineral Resource estimation.

Integration & Ecosystem

6.1 API Specifications

Proposed REST API Design

```

openapi: 3.0.0
info:
  title: Geological AI Platform API
  version: 1.0.0
  description: |
    RESTful API for AI-driven geological modeling.
    All endpoints require Bearer token authentication.

paths:
  /v1/projects:
    post:
      summary: Create new geological modeling project
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              required: [name, jurisdiction, commodity_type]
              properties:
                name: { type: string, example: "Pilbara Iron Project" }
                jurisdiction: { type: string, enum: [JORC, NI_43_101, SAMREC, PERC, CBRR] }
                commodity_type: { type: string, example: "iron_ore" }
                crs: { type: string, example: "EPSG:28350" }
      responses:
        '201':
          description: Project created
          content:
            application/json:

```

```

      schema:
        type: object
        properties:
          project_id: { type: string, format: uuid }
          created_at: { type: string, format: date-time }

/v1/projects/{project_id}/data/drillholes:
  post:
    summary: Upload drillhole data
    requestBody:
      required: true
      content:
        multipart/form-data:
          schema:
            type: object
            properties:
              file: { type: string, format: binary, description: "CSV/CSVZ/OMF format" }
              data_format: { type: string, enum: [csv, csvz, omf, deswik] }
    responses:
      '202':
        description: Data upload accepted for processing
        content:
          application/json:
            schema:
              type: object
              properties:
                upload_id: { type: string, format: uuid }
                status: { type: string, enum: [pending, processing, completed, failed] }

/v1/projects/{project_id}/models:
  post:
    summary: Generate 3D geological model
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            required: [model_type, parameters]
            properties:
              model_type:
                type: string
                enum: [pinns, diffusion, gnn, ensemble]
              parameters:
                type: object
                properties:
                  resolution: { type: integer, example: 256 }
                  physics_constraints: { type: boolean, default: true }
                  uncertainty_quantification: { type: boolean, default: true }
    responses:
      '202':

```

```

    description: Model generation queued
    content:
      application/json:
        schema:
          type: object
          properties:
            model_id: { type: string, format: uuid }
            status: { type: string, enum: [queued, running, completed, failed] }
            estimated_completion: { type: string, format: date-time }

/v1/models/{model_id}/export:
  get:
    summary: Export model in standard format
    parameters:
      - name: format
        in: query
        schema:
          type: string
          enum: [omf, gocad, leapfrog, vtk, csv]
    responses:
      '200':
        description: Model file
        content:
          application/octet-stream:
            schema:
              type: string
              format: binary

/v1/models/{model_id}/lineage:
  get:
    summary: Retrieve full data and model lineage
    responses:
      '200':
        description: Lineage graph
        content:
          application/json:
            schema:
              type: object
              properties:
                model_version: { type: string }
                training_data_version: { type: string }
                drill_programs: { type: array, items: { type: string } }
                model_parameters_hash: { type: string }
                created_by: { type: string }
                created_at: { type: string, format: date-time }

```

Authentication: OAuth 2.0 + JWT; role-based access control (viewer, modeler, admin, QP) **Rate Limiting:** 100 requests/minute for standard; 1000/minute for enterprise **Versioning:** URL path versioning (/v1/, /v2/); 12-month deprecation cycle

6.2 Data Lineage and Versioning

Critical for Regulatory Compliance

Provenance Model:

```
Model v3.2.1
├─ Trained on: Dataset v2.1.4
│   ├── Drill Program 2024-Q3 (v1.0.2)
│   │   ├── Holes DH-1001 to DH-1050
│   │   ├── Assay lab: ALS Chemex (certificate #2024-AL-4451)
│   │   └─ QA/QC: 5 blanks, 10 standards, 5 duplicates
│   ├── Drill Program 2024-Q2 (v1.0.1)
│   ├── Satellite imagery: Sentinel-2 (2024-06-15, tile T50HMK)
│   └─ Topography: SRTM 30m (public, v3.0)
├─ Model architecture: Diffusion-UNet v2.3
│   ├── Base checkpoint: Diffusion-Base v1.0 (trained on 10k synthetic models)
│   ├── Fine-tuning: 500 epochs on Dataset v2.1.4
│   └─ Hyperparameters: config_hash 0x8f3a2b...
├─ Validation: Cross-validation fold 3
│   ├── RMSE: 0.42% Cu
│   └─ MAE: 0.31% Cu
└─ Signed off by: QP Jane Smith (license #12345)
    └─ Date: 2024-12-15
```

Technical Implementation:

- **Data versioning:** DVC (Data Version Control) or LakeFS for large files
- **Model versioning:** MLflow or Weights & Biases with artifact tracking
- **Lineage graph:** Neo4j or Apache Atlas for relationship queries
- **Immutable audit log:** Append-only blockchain or signed hash chain
- **Export format:** W3C PROV standard for interoperability

6.3 MLOps for Geology

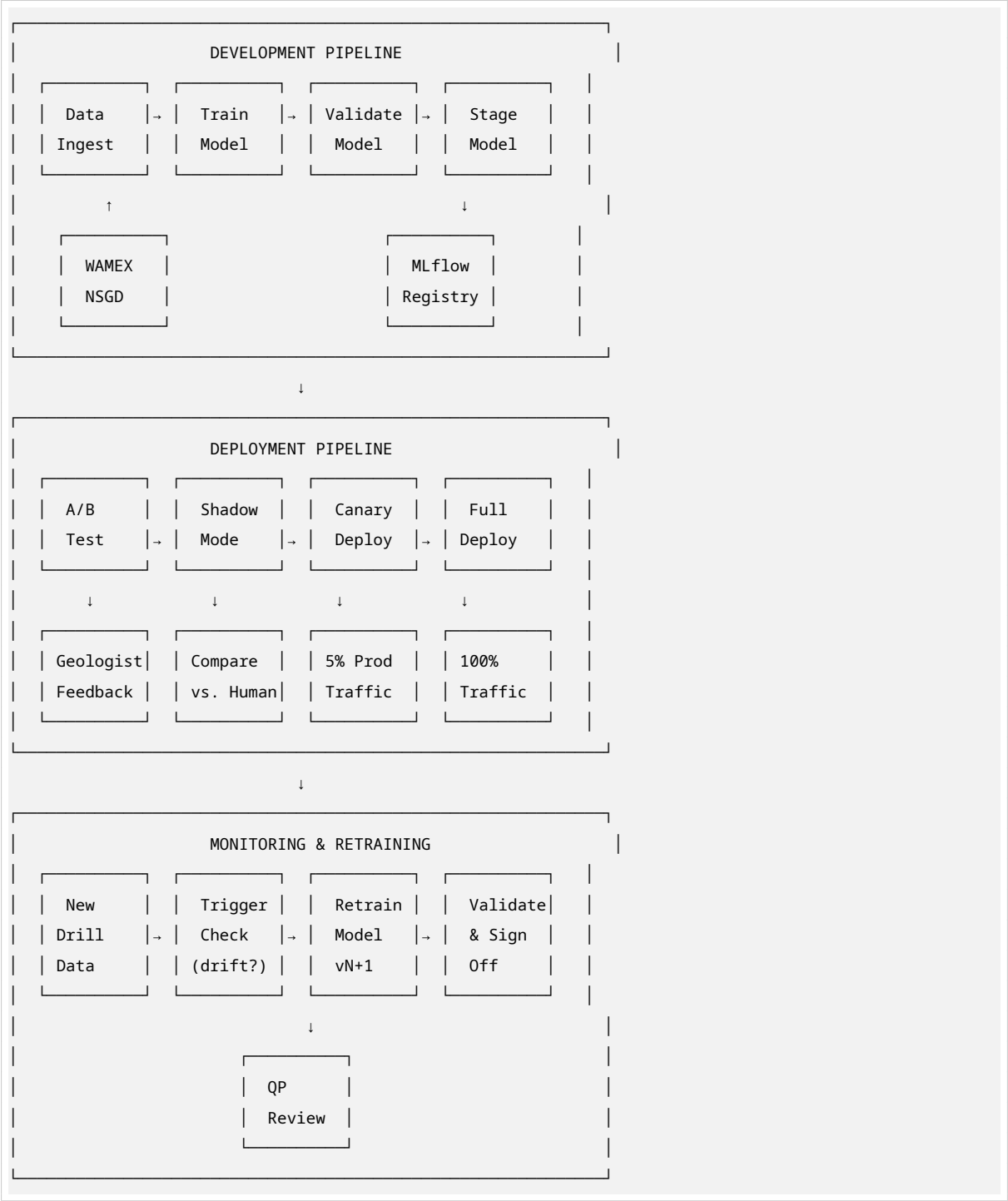
Unique Challenges vs. Consumer Tech MLOps

Challenge	Consumer Tech Approach	Geological Adaptation
Sparse data	Massive datasets, abundant labels	Few-shot learning; active learning; synthetic data generation
Expensive labels	Crowdsourcing, weak supervision	Expert-in-the-loop; transfer learning from synthetic geology
Non-stationary distributions	Concept drift monitoring	Geological domain shift (different terrane = different statistics)
Long feedback loops	Real-time A/B testing	Months/years until drill validation; proxy metrics needed
Regulatory compliance	Minimal	Full audit trail; QP sign-off; immutable records

Table 20 – continued

Challenge	Consumer Tech Approach	Geological Adaptation
Compute at edge	Cloud-only	Mine sites often offline; edge deployment required

Recommended MLOps Architecture:



Retraining Triggers:

- 1. New drill program completed (>10% increase in data volume)

- 2. **Concept drift detected** (distribution shift in new assays vs. training set)
- 3. **Quarterly schedule** (regardless of data volume)
- 4. **Model performance degradation** (proxy metric drift >10%)

 Visual & Interactive Assets

7.1 Interactive Technology Matrix

Concept: Filterable table where user selects data situation → recommended AI stack.

Filter Dimensions:

- Data availability: [Sparse drillholes / Dense drillholes / Seismic / Satellite / Multi-physics]
- Budget: [Low (<\$50K) / Medium (\$50K-\$500K) / High (>\$500K)]
- Timeline: [Rapid (<1 month) / Standard (1-6 months) / Extended (>6 months)]
- Compliance requirement: [None / Internal / JORC / NI 43-101 / SAMREC]
- Commodity: [Generic / Porphyry Cu / Orogenic Au / Lithium / etc.]

Output:

- Recommended primary technology
- Recommended secondary/ensemble approach
- Expected accuracy range
- Estimated compute cost
- Key risks and mitigations

Implementation: React/Vue component with state-based filtering; embed in documentation site.

7.2 Decision Tree Flowchart

```
START: What is your data situation?
|
├─ I have only sparse drillholes (<50 holes)
|   ├── With geochemistry only → GNN + Kriging ensemble
|   ├── With geophysics coverage → PINN-constrained inversion
|   └─ With satellite imagery → Multi-modal fusion (GNN + CNN)
|
├─ I have dense drillholes (>500 holes)
|   ├── Need rapid updates → Diffusion model infill
|   ├── Need uncertainty quantification → Ensemble of PINNs
|   └─ Need structural geology → Transformer-based sequence model
|
└─ I have seismic data
    ├── 2D seismic lines → CNN segmentation + interpretation
    ├── 3D seismic volume → 3D U-Net + geophysical inversion
    └─ Time-lapse (4D) → LSTM/Transformer temporal model
```

```
|
├─ I have remote sensing only
|   ├── Regional prospectivity → Random Forest / XGBoost
|   ├── Alteration mapping → CNN on Sentinel-2 / ASTER
|   └─ Structural mapping → U-Net on DEM + SAR
|
└─ I need JORC/NI 43-101 compliance
    ├── Assistive AI (human-in-the-loop) → All technologies with audit trail
    └─ Autonomous AI → NOT RECOMMENDED for current regulations
```

7.3 Comparison Tables

Platform Feature Comparison

Feature	Leapfrog Geo	Maptek Vulcan	VRIFY DORA	Micromine	GeologicAI	[Your Platform]
Pricing Model	Daily/Sub	Quote	Quote	Quote	Quote	SaaS/Usage
Pricing (indicative)	\$156/day	Undisclosed	Undisclosed	Undisclosed	Undisclosed	TBD
3D Modeling	✔ Implicit	✔ Explicit	✘ Target gen	✔ Explicit	✘ Scanning	✔ AI-gen
AI/ML Features	🟡 Emerging	✘ None	✔ Core	✘ None	✔ Core imagery	✔ Core
Data Formats	CSV, OMF, DXF	Vulcan native	Proprietary	CSV, OMF	Core photos	OMF, CSV, VTK
Cloud API	Evo (new)	✘ On-prem	✔ Cloud	✘ Desktop	✔ Cloud	✔ Cloud
JORC/NI 43-101	✔ Evo API	🟡 Limited	✘ None	✘ None	✘ None	✔ Full REST
Export Options	✔ Industry std	✔ Industry std	🟡 Unclear	✔ Industry std	✘ N/A	🟡 Building
Support Quality	OMF, VTK, CSV	Vulcan, CSV	PDF, Web	OMF, CSV	PDF, Images	OMF, VTK, CSV
On-premise Option						TBD
	✔	✔	✘	✔	✘	✔ Enterprise

🎯 Priority Roadmap

Immediate Actions (Next 30 Days)

Priority	Action	Owner	Deliverable
 1	Compile real pricing data via RFQs	Business	Pricing intelligence matrix
 2	Search SEDAR/ASX for AI in NI 43-101/JORC	Research	Regulatory precedent report
 3	Design WAMEX benchmark protocol	Technical	Benchmark specification doc
 4	Interview 5 working geologists	Product	Customer insight synthesis

Short-Term (Next 90 Days)

Priority	Action	Owner	Deliverable
 5	Execute WAMEX benchmark on 5 reports	Technical	Reproducible results + leaderboard
 6	Draft API specification v0.1	Engineering	OpenAPI YAML + documentation
 7	Build commodity-specific model (Porphyry Cu)	Technical	Trained model + evaluation
 8	Develop financial model v1.0	Business	Spreadsheet with assumptions

Medium-Term (Next 6 Months)

Priority	Action	Owner	Deliverable
 9	Run blind test protocol with 20 geologists	Research	Trust assessment report
 10	Complete economic validation case study	Business	ROI analysis for 1 deposit
 11	Implement data lineage system	Engineering	Provenance tracking demo
 12	Develop MLOps pipeline for geology	Engineering	CI/CD for model retraining

Highest-Value Additions (Ranked)

1. **Real pricing data** from existing vendors — *Hard to get but transformative for business case validation*
2. **Regulatory precedent** — *Even one example of an AI-assisted report accepted by a regulator would de-risk the entire market*

- 3. **Reproducible benchmark** on an open geological dataset — *Establishes credibility and technical differentiation*
- 4. **Customer interview synthesis** — *Even 5-10 conversations with working geologists would validate product-market fit*

Appendix A: Data Sources and References

Search Results Summary

Query	Results	Key Finding
Leapfrog Geo pricing 2025	✔ Found	Consultants Daily: \$156/day, cap \$2,340/month
Maptek Vulcan pricing	⚠ Partial	Quote-based only; no public pricing
VRIFY pricing	✖ None	\$12.5M Series B funding confirmed; pricing undisclosed
Seequent AI integration	✔ Found	Evo platform launched 2025; AI features at PDAC 2026
AI in JORC/NI 43-101	⚠ Partial	No confirmed AI-assisted report acceptance found
WAMEX dataset	✔ Found	120K reports; 105K public; 5-year confidentiality
NSGD Nevada	✖ None	No specific open dataset identified

Key URLs

- Seequent Pricing: <https://www.seequent.com/help-support/seequent-id/consultants-daily/>
- Seequent Evo Announcement: <https://www.bentley.com/news/introducing-seequent-evo/>
- Seequent PDAC 2026: <https://www.seequent.com/seequent-brings-ai-and-cloud-enabled-geoscience-to-pdac-2026/>
- WAMEX Portal: <https://wamex.dmp.wa.gov.au/Wamex/Home/HomePage>
- VRIFY Funding: <https://ca.finance.yahoo.com/news/vrify-lands-12-5m-series-150000659.html>
- NI 43-101 Standards: <https://mrmr.cim.org/media/1017/national-instrument-43-101.pdf>
- JORC/NI 43-101 Comparison: <https://www.miningdoc.tech/question/what-are-jorc-ni-43-101-what-are-they-for/>

Appendix B: Glossary

Term	Definition
PINN	Physics-Informed Neural Network —neural network trained with physical law constraints
GNN	Graph Neural Network —operates on graph-structured data (e.g., drillhole networks)
Diffusion Model	Generative model that learns to reverse a noise-corruption process
NeRF	Neural Radiance Field —implicit 3D scene representation

Table 26 – continued

Term	Definition
QP	Qualified Person –NI 43-101 requirement for technical report authorship
CP	Competent Person –JORC Code equivalent of QP
WAMEX	Western Australian Mineral Exploration reports database
NSGD	Nevada State Geothermal Database (proposed; requires verification)
TAM/SAM	Total Addressable Market / Serviceable Addressable Market
LTV/CAC	Lifetime Value / Customer Acquisition Cost
MLOps	Machine Learning Operations –DevOps practices for ML systems
DVC	Data Version Control –Git-like system for data and models

Appendix C: Contributing Guidelines

To contribute to this research repository:

1. **Fork** the repository and create a feature branch
2. **Add** your research in the appropriate section using the established format
3. **Cite** all sources with inline citations and add to Appendix A
4. **Update** the Priority Roadmap if your contribution changes priorities
5. **Submit** a pull request with a clear description of additions

Research Quality Standards:

- All pricing claims must include source and date
- All technical benchmarks must be reproducible (code + data)
- All regulatory claims must reference specific filings or official guidance
- All customer insights must be anonymized and include sample size

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